MUHAMMAD HAROON RASHID

Full Stack Developer (Generative AI)
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SUMMARY

Full Stack Developer specializing in Generative AI applications, LLM integrations and scalable Python backends (FastAPI). Experienced in building RAG pipelines, embeddings-based search, AI agents (LangChain, LangGraph) and REST APIs, and deploying them end to end with Git, Docker and cloud platforms. Strong problem-solving skills, clear communication, a team-oriented mindset and a self-driven approach to learning new AI technologies. Open to full-time remote roles.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, HTML, CSS, C++, C#
- **Backend and Automation:** FastAPI, REST APIs, Webhooks, MCP (FastMCP), n8n
- **Generative AI:** Large Language Models (LLMs), Embeddings, AI Integrations, LLM Agents and Tool Calling, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, Hugging Face
- **Machine Learning and Data:** Machine Learning, scikit-learn, Exploratory Data Analysis (EDA), Pandas, NumPy, Matplotlib, Seaborn
- **Databases:** PostgreSQL, Oracle, SQLite, Supabase, pgvector, FAISS (vector database)
- **Frontend:** Next.js, React, Tailwind CSS, Streamlit
- **Deployment and Tools:** Git, GitHub, Docker, Vercel, Render
- **AI-Assisted Development:** Claude Code (Skills, Subagents, Plugins, Hooks, Slash Commands, MCP, Context Window Management), Antigravity
- **Soft Skills:** Problem-solving, communication, teamwork, curiosity, self-driven, time management

EXPERIENCE

AI Automation and Full Stack Developer | IES Group (Engineering Firm) | 2026 - Present

- Designed and deployed an AI-powered lead routing system (n8n Cloud, Groq LLM, Zoho SMTP) that classifies incoming inquiries and routes them to the right department across four departments.
- Built and deployed the company web platform (GitHub to Vercel) and configured professional email infrastructure (Zoho Mail: MX, SPF, DKIM) for four department addresses.
- Built the EPADS tender intelligence system and a WhatsApp assistant (Python, FastMCP, SQLite) to support government tender bidding.
- Worked directly with business stakeholders to turn manual processes into automated, deployed AI solutions.

PROJECTS

TubeMind AI: RAG-Based YouTube Video Chat Assistant | Python, LangChain, FastAPI, FAISS, Hugging Face, Streamlit

- Built a Retrieval-Augmented Generation (RAG) pipeline with transcript extraction, chunking, Hugging Face embeddings and FAISS vector search, using Groq-hosted Llama 3.3 70B LLM with streaming responses.
- Developed FastAPI endpoints (/ingest, /chat, /chat_stream), diagnosed and fixed a production retrieval issue causing stale context reuse, and restructured the code into a modular backend/frontend architecture.

EPADS Tender Intelligence System and WhatsApp Assistant | Python, FastMCP, SQLite

- Automated searching and reading of tenders on Pakistan's EPADS portal: crawler and parsers for tender listings, details and bidding documents, with SQLite storage and change detection for updated tenders.
- Built a FastMCP server (search_tenders, get_tender_details) so AI assistants can search and read tenders, plus a daily digest CLI with scheduler.
- Integrated a WhatsApp chatbot that returns a tender summary card (agency, title, closing date, bid security, required registrations) with export to Excel, PDF and Word.
- Live-tested against epads.gov.pk and covered by an automated test suite (31 tests).

HostelDesk: AI-Powered Hostel Complaint System | n8n, PostgreSQL, FastAPI, Next.js

- Built a full-stack system where students report issues conversationally through Slack; an n8n AI Agent (LLM with chat memory) creates complaints in PostgreSQL, auto-assigns the correct Resident Tutor by hostel, logs status history and sends real-time Slack notifications.
- Developed a FastAPI webhook backend and a Next.js 14 staff dashboard with filters and workload views; deployed on Vercel and Render. Designed the PostgreSQL schema for students, complaints, hostels, staff, roles and status history.

Other Projects: AgriScribe (GenAI crop-disease assistant, UETIANS / HEC Pakistan / Pak Angels Gen AI Hackathon), Facial Emotion Detection, Phishing Website Detection (Machine Learning), House Price Predictor (Streamlit).

EDUCATION

BS Computer Science | University of Engineering and Technology (UET) Lahore | 2024 - 2028 (currently enrolled)
Intermediate (ICS) | BISE Sahiwal | 2024 | Grade A
Matric (Science) | BISE Sahiwal | 2021 | Grade A+

CERTIFICATIONS AND AWARDS

- Claude Code in Action, Anthropic Education (May 2026)
- Generative AI Application Developer, UETIANS / HEC Pakistan / Pak Angels (Top Performer)
- CM Laptop Award, Chief Minister Punjab (Mar 2025)
- Honhar Scholarship, Government of Punjab (Oct 2024)

LANGUAGES

Punjabi / Urdu (native), English